

Performance Analysis and Walltime Prediction for Neuroscience Applications

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Abstract—With the recent upcomming of HPC, BigData and AI, the need for computational resources is ever increasing in the field of neuroscience. However, efficiently utilizing these resources remains a challenge due to the complexity of modern computing systems and the diverse nature of neuroscience applications. In this paper, we present a comprehensive performance analysis of several neuroscience applications, focusing on their execution time and memory footprint. In a second step, we will develop a model to predict the execution time and memory usage of these applications based on their input parameters and system configurations.

Index Terms—Stochastic application, Execution time, Memory footprint

I. INTRODUCTION

REFERENCES